Retail Sales Analytics Report

Data Cleaning · Exploratory Analysis · Predictive Modeling · Dashboard

Prepared by: Data Analytics Team
Date: August 2026

1. Project Overview

This project delivers an end-to-end data analytics pipeline for a simulated multi-category retail business, covering data cleaning, exploratory data analysis, statistical analysis, correlation analysis, predictive modeling, and an interactive dashboard, packaged as a GitHub-ready repository.

2. Dataset Description

The dataset contains 2,015 order-level records spanning January 2022 to December 2024, across 6 product categories and 5 regions. The raw data intentionally contains missing values, duplicate rows, and outliers to demonstrate a realistic cleaning workflow.

Column	Description
OrderID	Unique order identifier
Date	Order date
Category / Product	Product taxonomy
Region	Sales region
UnitPrice, Quantity, DiscountPct	Transaction attributes
Sales, Cost, Profit	Financial outcomes
CustomerAge, SatisfactionScore	Customer attributes

3. Objectives

• Clean and prepare the raw dataset for analysis.
• Understand sales, profit, and customer patterns across categories, regions, and time.
• Quantify relationships between key numeric variables.
• Build predictive models for order value and high-value order classification.
• Present findings through visualizations, a written report, and an interactive dashboard.

4. Methodology

The pipeline (run_pipeline.py) executes six stages: preprocessing, exploratory analysis, visualization, modeling, prediction export, and dashboard delivery. See README.md for the full technical breakdown of each stage.

5. Data Cleaning

Step	Result
Duplicate rows removed	15 exact duplicates dropped
Missing values imputed	Numeric via median, categorical via mode
Outliers capped (IQR)	915 extreme values capped
Features engineered	Year, Month, Quarter, ProfitMargin, AvgOrderValue, HighValueOrder

6. Exploratory Data Analysis

Metric	Value
Total Sales	$677,423
Total Profit	$218,695
Average Order Value	$338.71
Average Profit Margin	32.4%

Category	Total Sales
Electronics	$283,316
Home & Kitchen	$121,492
Sports	$92,190
Clothing	$69,312
Toys	$61,749
Beauty	$49,364

7. Visualizations

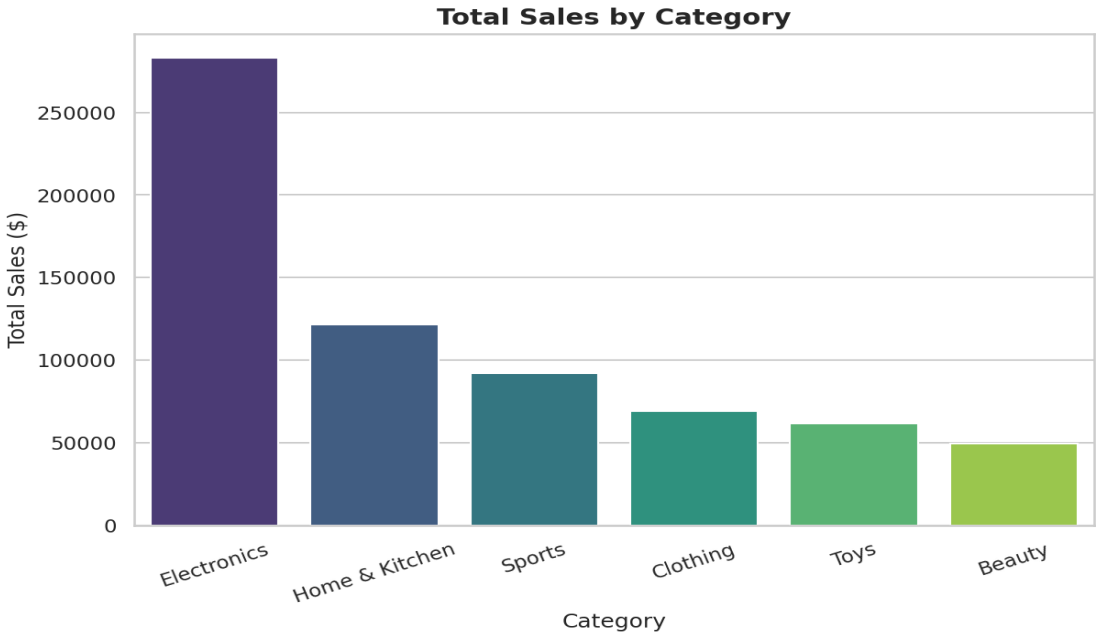


Figure 1: Total Sales by Category

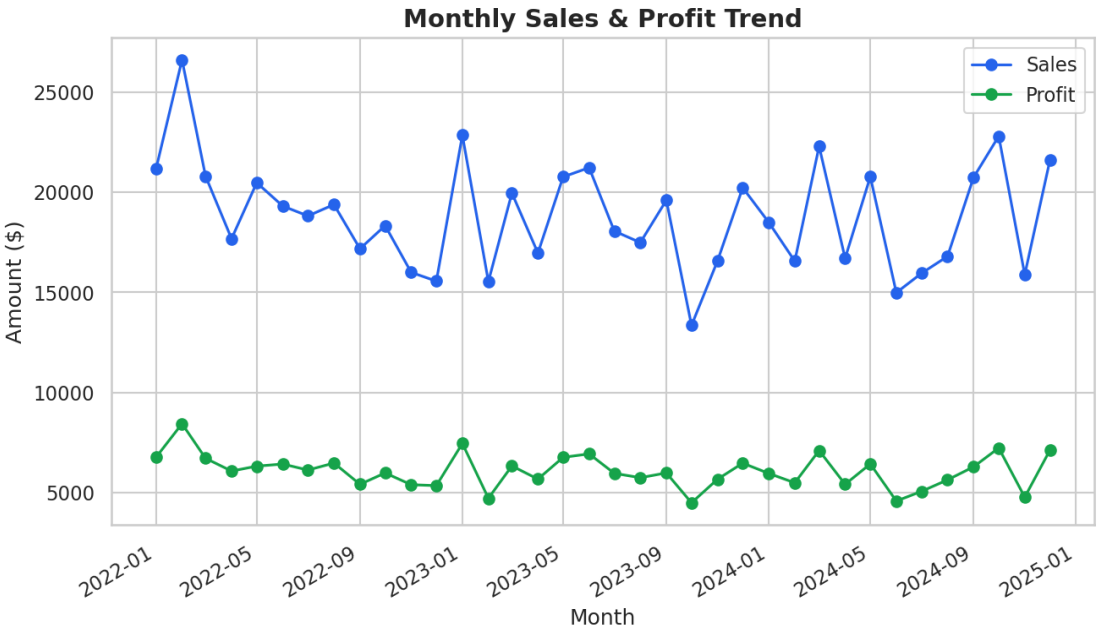


Figure 2: Monthly Sales & Profit Trend

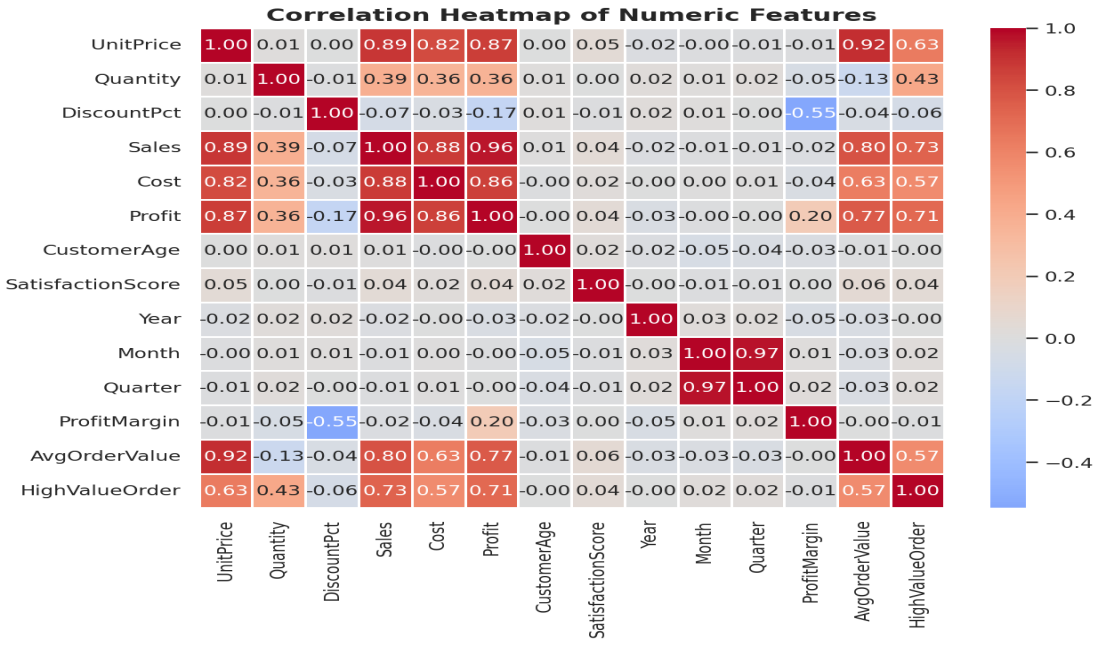


Figure 3: Correlation Heatmap

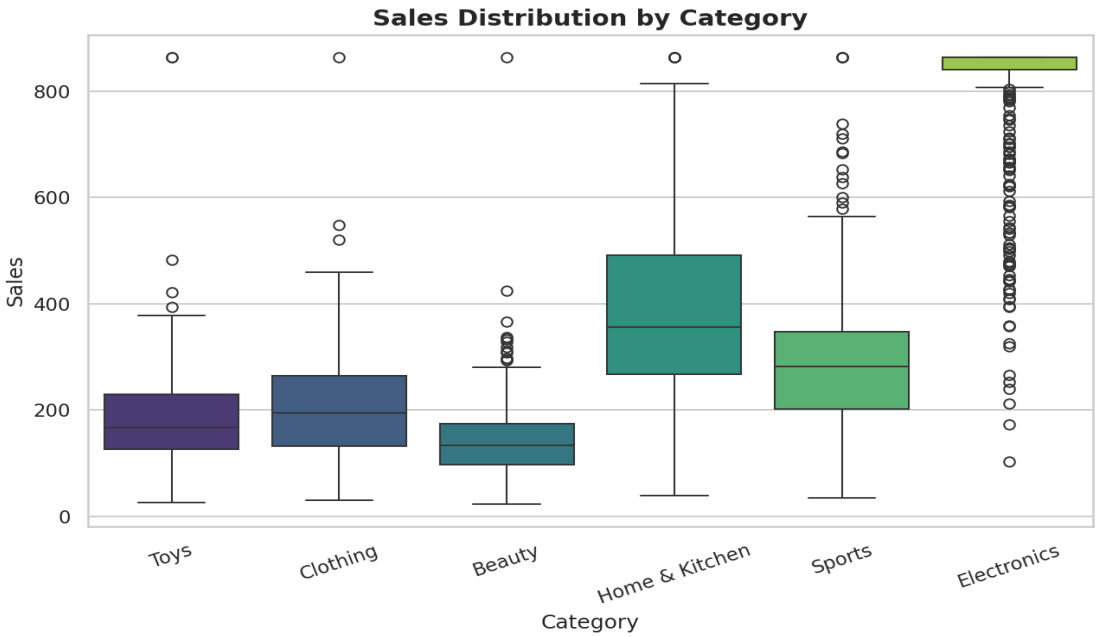


Figure 4: Sales Distribution by Category

8. Statistical Findings

Sales values are right-skewed (skew ≈ 1.09), consistent with a long tail of large orders. UnitPrice and Sales show a positive correlation, as expected. SatisfactionScore shows negligible correlation with Sales or Profit in this simulated dataset. Profit margin is fairly stable across categories, averaging around 32%.

9. Machine Learning

9.1 Regression — Predicting Order Sales

Metric	Value
MAE	18.73
RMSE	44.09
R^2	0.971

9.2 Classification — High-Value Order Detection

Metric	Value
Accuracy	0.980
Precision	0.980
Recall	0.980
F1 Score	0.980
ROC AUC	0.998

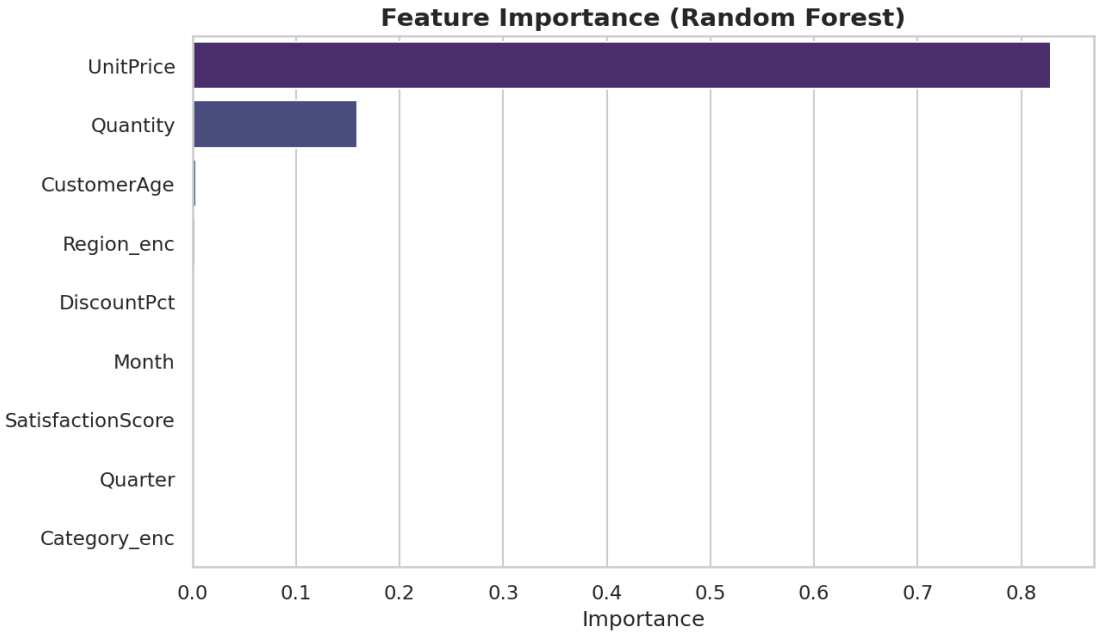


Figure 5: Feature Importance (Regression Model)

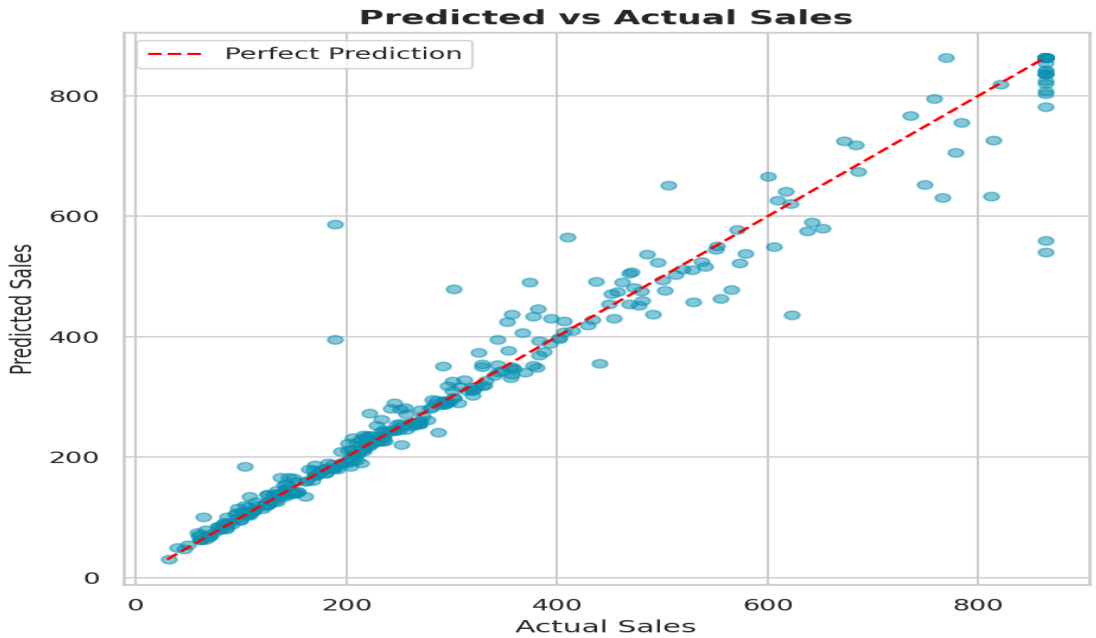


Figure 6: Predicted vs. Actual Sales

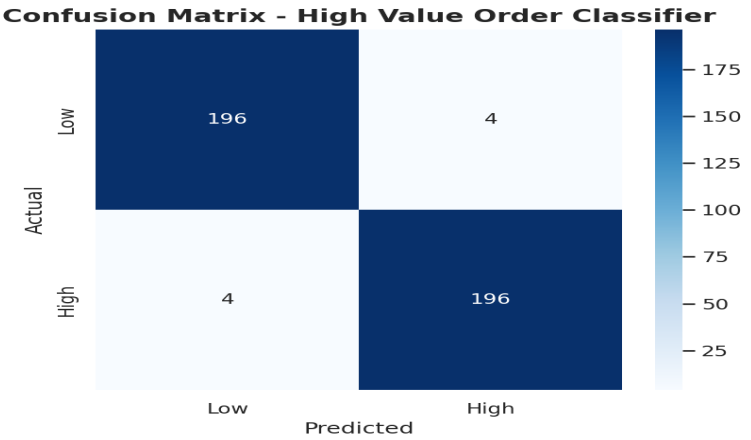


Figure 7: Confusion Matrix

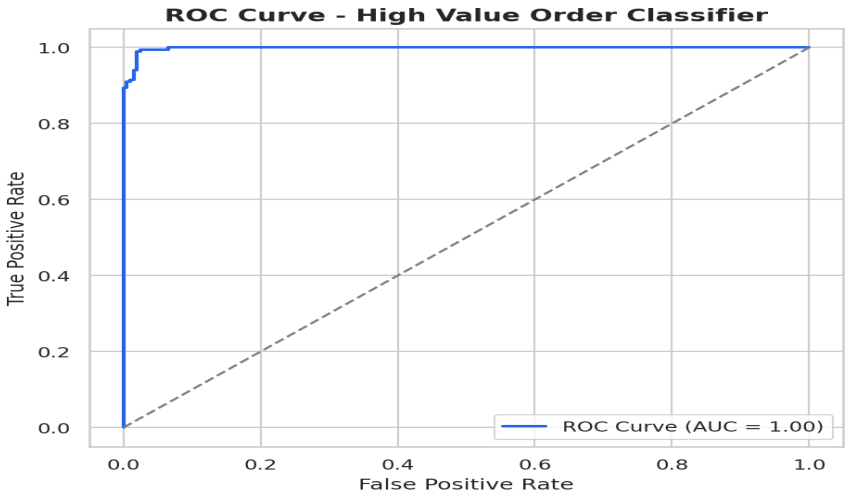


Figure 8: ROC Curve

10. Model Evaluation

Both models perform strongly, which is expected given Sales is partly deterministic from UnitPrice, Quantity, and DiscountPct in this simulated dataset. The residual plot shows residuals centered near zero with no strong pattern. On real-world data, expect noisier relationships and lower performance; these results should be treated as a template methodology.

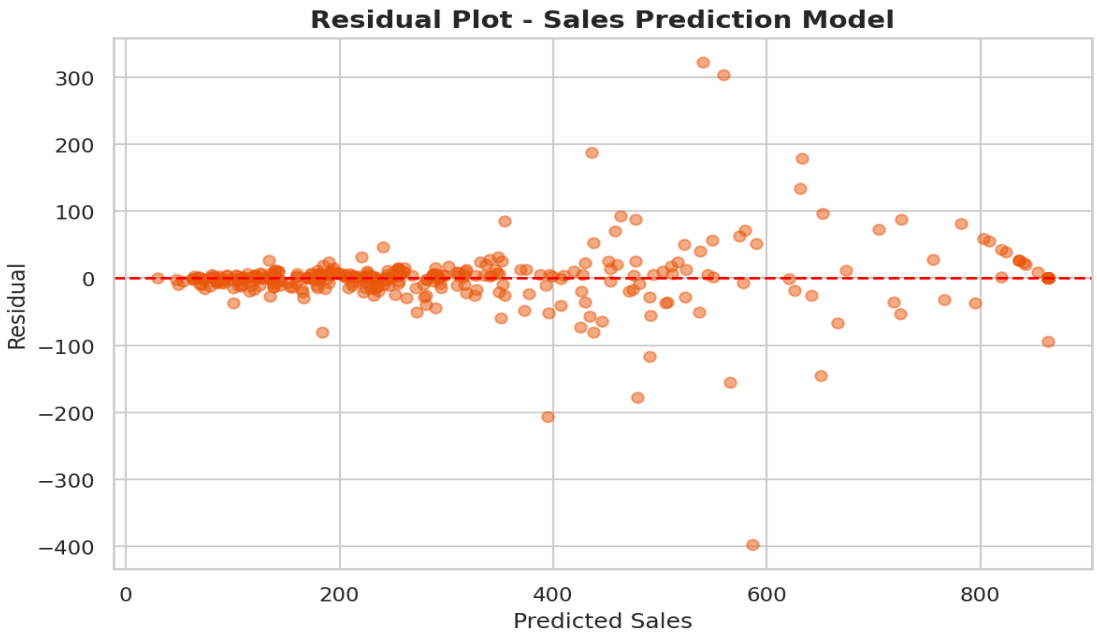


Figure 9: Residual Plot

11. Business Insights

• Electronics is the primary revenue driver and merits continued investment.
• Regional performance is balanced; growth efforts may favor cross-region initiatives.
• Discounting is used sparingly (mean ≈ 3.3%) and could be tested further.
• Profit margin is consistent across categories, indicating balanced pricing.

12. Recommendations

• Expand Electronics assortment given strong average order value.
• Run targeted discount A/B tests in Beauty and Toys.
• Use the HighValueOrder classifier to prioritize retention campaigns.
• Monitor SatisfactionScore alongside NPS or churn data in production.

13. Future Improvements

• Incorporate real transactional data and seasonality effects.
• Add time-series specific forecasting models (Prophet, ARIMA).
• Add customer-level features such as lifetime value.
• Deploy the dashboard with authentication and scheduled data refresh.

14. Conclusion

This project demonstrates a complete, reproducible retail analytics workflow — from raw, messy data to cleaned datasets, rich visualizations, validated predictive models, and an interactive dashboard. The modular src/ package and run_pipeline.py entry point make it straightforward to swap in a real dataset and reproduce the full analysis.